

Topics in Applied Graph Theory

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based on course by Priyavrat Deshpande

Evaluation: Assignments + Projects

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1. Introduction

Definition: Graph

A graph $G = (V, E)$ is specified by two finite sets V and E where V is the set of vertices and E is the set of edges (a subset of 2 subsets of V).

Definition: Directed Graph

A directed graph $G = (V, E)$ is specified by two finite sets V and E where V is the set of vertices and E is the set of edges (a subset of $V \times V$ barring (x, x) or loops).

Side Note

The graph is simple and undirected (notice 2 subsets) while a directed graph is, well, directed (notice the cartesian product).

Definition: Multigraph

A multigraph G has E be a multiset.

Definition: Weighted

We say a graph is (edge) weighted, if we have a function $w : E \rightarrow \mathbb{R}$

For notational convenience, we take $V = [n]$.

Definition: Adjacency Matrix

Adjacency matrix of graph G , called A_G is a $n \times n$ matrix where $a_{i,j} = 1 \iff (i,j) \in E$ and 0 otherwise.

In weighted graph, $a_{i,j} = w((i,j))$

Definition: Degree

The degree of vertex i is number of edges containing said vertex, that is $d_G(i) = \deg(i) = \sum_{j=1}^n a_{i,j}$.

Definition: Degree Matrix

The degree matrix of D is a diagonal matrix $D(i,i) = d(i)$.

Definition: Laplacian

The Laplacian of graph G is $\mathcal{L} = D - A$

Definition: Graph Energy

For $x \in \mathbb{R}^n$, $x^T \mathcal{L} x$ is called graph energy.

Example

Given a 2 path, $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \mathcal{L} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \Rightarrow x^T \mathcal{L} x = (x_1 - x_2)^2 \geq 0$.

Similarly, for a 3 path, $x^T \mathcal{L} x = (x_1 - x_2)^2 + (x_2 - x_3)^2$

Theorem 1.1. For a weighted graph,

$$x^T \mathcal{L} x = \sum_{(i,j) \in E} w_{i,j} (x_i - x_j)^2$$

Proof.

$$\begin{aligned}
x^T \mathcal{L}x &= x^T(D - W)x \\
&= x^T Dx - x^T Wx \\
&= \sum d_i x_i^2 - \left(\sum w_{i,j} x_i x_j \right) \\
&= \sum d_i x_i^2 - 2 \sum_{(i,j) \in E} w_{i,j} x_i x_j \\
&= \sum_{(i,j) \in E} (w_{i,j} x_i^2 - 2w_{i,j} x_i x_j + w_{i,j} x_j^2) \\
&= \sum_{(i,j) \in E} w_{i,j} (x_i - x_j)^2
\end{aligned}$$

■

Corollary 1.2. $\mathcal{L}$ is Symmetric and Positive Semi-Definite.

This implies that $\mathcal{L}$ has non-negative real Eigenvalues.

Definition: Spectrum of Laplacian

We define $\sigma(\mathcal{L}) := \{0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n\}$ as the spectrum of $\mathcal{L}$

Let $\mathbf{1} = (1, 1, \dots, 1)^T$. Notice,

$$\begin{aligned}
\mathcal{L}\mathbf{1} &= (D - W) \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= D \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} - W \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} d_1 \\ \dots \\ d_n \end{pmatrix} - \begin{pmatrix} \sum w_{1,j} \\ \dots \\ \sum w_{n,j} \end{pmatrix} \\
&= 0
\end{aligned}$$

which implies $\lambda_1(\mathcal{L}) = 0$ with eigenvector $\mathbf{1}$.

Theorem 1.3. $\lambda_2 > 0 \iff G$ is connected

Proof. ($\implies$) Assume $G = G_1 \sqcup G_2$ which implies $\mathcal{L}_G = \begin{pmatrix} \mathcal{L}_{G_1} & 0 \\ 0 & \mathcal{L}_{G_2} \end{pmatrix}$.

This implies we have atleast 2 independent vectors: $\begin{pmatrix} \mathbf{1}_{G_1} \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ \mathbf{1}_{G_2} \end{pmatrix}$ which have eigenvalue 0 which makes $\lambda_2 = 0$.

($\impliedby$) Suppose G is connected. Let $y \in \ker(\mathcal{L}_G)$.

$$\begin{aligned}\Rightarrow y^T \mathcal{L}y &= \sum_{(i,j) \in E} (x_i - x_j)^2 \\ &= 0\end{aligned}$$

This implies that $x_i = x_j \iff (i, j) \in E$. Thus, if vertex i and j are connected, then by a series of equalities, we'll end up with $y = \mathbb{1}$.

Thus, $\lambda_2 > 0$. ■

Corollary 1.4. *The number of connected components of G is equal to the multiplicity of 0 as an eigenvalue of L .*

We could define the multiplicity of 0 as k in $\chi_{\mathcal{L}}(t) = t^k q(t)$ where $t \nmid q$.

Example : Eigenvalues of complete graph

$G = K_n$ (the complete graph)

What are the eigenvalue?

Solution(Mine).

$$\begin{aligned}\mathcal{L}x &= \lambda x \\ \Rightarrow (n-1)x_i - \sum_{j \neq i} x_j &= \lambda x_i \\ \Rightarrow nx_i - \sum_j x_j &= \lambda x_i \\ \Rightarrow \text{either } x_1 = x_2 = \dots = x_n \text{ or } \lambda &= n\end{aligned}$$

As it is connected, the second eigenvalue can't be 0. Thus, all other eigenvalues are n . ■

Prof. Priyavrat's Solution. As we have n eigenvectors, then they form a eigenbasis.

Thus, for eigenvector y , $\langle y, \mathbb{1} \rangle = \sum_j y_j = 0$.

$$\begin{aligned}\Rightarrow (\mathcal{L}y)_i &= ny_i - \sum_j y_j \\ &= ny_i \\ \Rightarrow \mathcal{L}y &= ny\end{aligned}$$

Thus, n is an eigenvalue with multiplicity $n - 1$. ■

Example : Eigenvalues of Star Graph

Given $G = S_n$ (the star graph), What are the eigenvalue?

Solution.

TODO.



2. Laplacian Eigenmaps

While we have ‘barely scratched the surface’ of spectral graph theory, we already have an application.

In python’s Scikit Learn’s manifold learning we have a class **spectral embedding**. This is done by Belkiss-Niyogi process (found in 2000).

Problem 2.1. Given data (a finite set of vectors) $\subseteq \mathbb{R}^{\text{high}}$. find $\Phi : \mathbb{R}^{\text{high}} \rightarrow \mathbb{R}^{\text{low}}$ preserving some property.

For example PCA preserves variance while LLE preserves neighborhoods.

The high-level flowchart is:

Data $\rightarrow$ Data Graph (weighted to allow discrimination) $\rightarrow$ Some eigenvalues of L (equal to low) $\rightarrow$ extract coordinates from these eigenvectors.

Note, this doesn’t preserve the original coordinates as we move from data to an abstract graph.

TODO. *Example*

How do we find such coordinates?

high $w_{i,j} \Rightarrow$ low $|x_i - x_j| \equiv$ low $(x_i - x_j)^2$
 low $w_{i,j} \Rightarrow$ high $|x_i - x_j| \equiv$ high $(x_i - x_j)^2$

This intuition can be formalized/translated to given optimization problem:

$$\min_{x \in \mathbb{R}^n} \sum_{i,j} w_{i,j} (x_i - x_j)^2$$

which is nothing but $x^T \mathcal{L} x$. To prevent $x_i = 0$ type solutions and simply scaled solutions, we can but $\|x\| = 1$.

Also, to prevent $\mathbf{1}$ (appropriately) scaled from returning as solution, we can just add $x \in \mathbf{1}^\perp$.